SF-Mamba: Rethinking State Space Model for Vision

Masakazu Yoshimura[1], Teruaki Hayashi[2], Yuki Hoshino[1], Wei-Yao Wang[1], and Takeshi Ohashi[1] (Sony Group Corporation[1], Sony Interactive Entertainment[2])

The code is available!

Introduction

Mamba is theoretically efficient than Attention

Attention FLOPs: $O(LD^2 + L^2D)$ L: token length

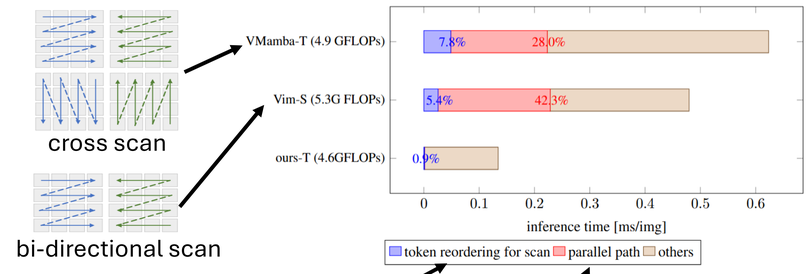
Mamba FLOPs: $O(LD^2 + LD)$ D: channel dimension

But Vision Mamba tend to be slow in processing low resolution images…

	MACs	224x224 Throughput [img/s]	
Deit-S (Attention)	4.6G	4700	similar MACs
Vim-S (Mamba)	5.3G	1100	but much slower

Our Investigation

Well-used multi-directional scan incurs the speed



Token reordering cost is not negligible

Parallel path of course double the cost of SSM part

Mamba's parallel scan is sub-optimal for short sequence

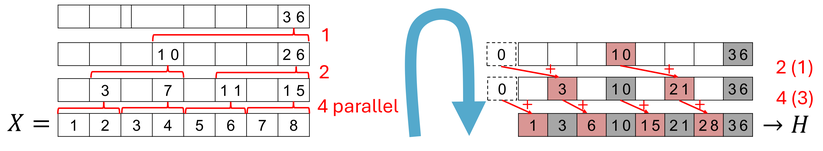
Original recurrent computation: $H_t = A_t H_{t-1} + B_t X_t$

e.g. $A = B = 1$ and $X =$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

1+2=3, 3+3= 6, 6+4=10, … (compute sequentially step by step)

$H =$ | 1 | 3 | 6 | 10 | 15 | 21 | 28 | 36 |

Parallel scan:



GPU hardware constraint: Working with groups of 32 (64) threads

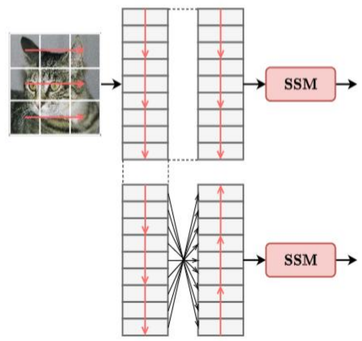
→ **Launching 32 threads to compute 4/2/1 operations**

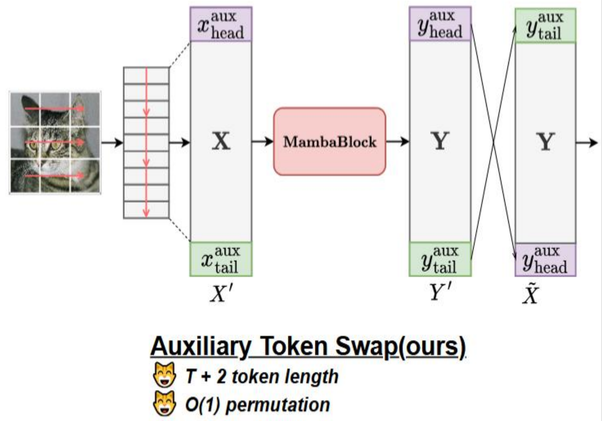
Methodology

Rethinking Vision Mamba from Data Flow Perspective

Auxiliary token swapping with unidirectional scan

Two auxiliary tokens are introduced, and by swapping them between Mamba blocks, future-to-past information flow is achieved under unidirectional scan

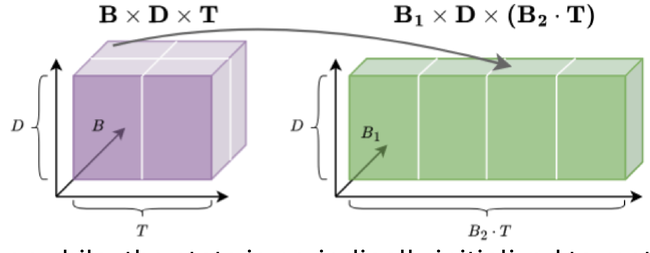




Rethinking Vision Mamba from Computational Perspective

Batch folding with periodic state reset

Because Mamba's efficiency only holds for long token lengths, we move part of the batch dimension into the token dimension to artificially increase the effective sequence length



Meanwhile, the state is periodically initialized to match the original computation

Periodic State Reset Trick

for $t = 1$ **to** $B_2 \cdot T$
  if $t \bmod T = 0$
    $A_t \leftarrow 0$
  $h_t \leftarrow A_t h_{t-1} + B_t x_t$
  $y_t \leftarrow C_t h_t$

$$H_t = A_t H_{t-1} + B_t x_t$$

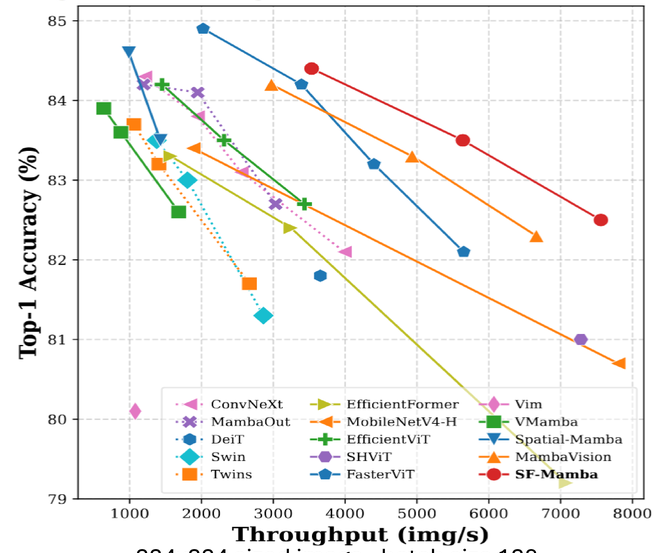
$$Y_t = C_t H_t$$

Evaluation

Experimental Setup

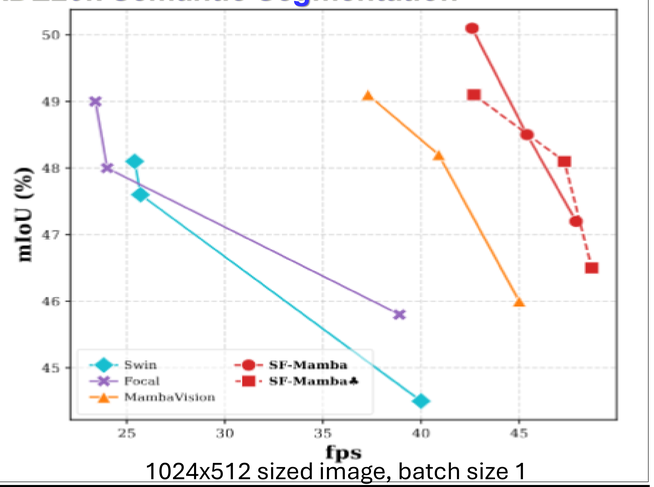
We implement our two proposals in Mamba block of MambaVision (CVPR25) architecture

ImageNet-1K Image Classification



224x224 sized image , batch size 128

ADE20k Semantic Segmentation



1024x512 sized image, batch size 1

Conclusion

Proposed **auxiliary token swapping** and **batch folding** offer good accuracy-efficiency trade-off.